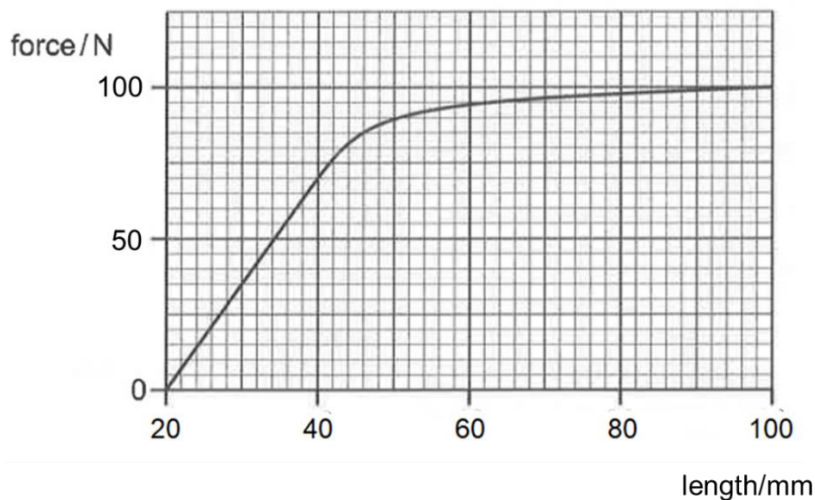


7

A rubber strip is stretched by a force. The graph of the applied force against its length is shown.



What is the work done in stretching the strip to the first point where it no longer obeys Hooke's Law?

A 0.70 J

B 6.3 J

C 700 J

D 6300 J

Solution: A

The limit of proportionality is when the strip no longer obeys Hooke's Law. This occurs at 40mm of length (20 mm of extension).

Work done

= area under the graph up to 40mm

= $\frac{1}{2} (0.0020 \text{ m})(70\text{N})$

= 0.70 J

8

A ribbon, wound around a reel, is pulled and rotates about its centre. The ribbon leaves the